Ayushi Agrawal

+1 (209) 279-6913 | ayushiamitagrawal@gmail.com | www.linkedin.com/in/ayushiaagrawal | https://github.com/ayushi8795

Summary

Software Developer with 5 years of strong expertise in Python and hands-on experience building ETL workflows and data transformation processes. Skilled in creating automation scripts, backend services, and SQL-based data modeling to support analytics and reporting needs. Worked in Agile environments and efficiently led Java project, improving system performance, stability, and user experience. Known for debugging complex issues, optimizing application logic, and reducing manual work through well-designed automation. Seeking a role that combines Python development with data-focused engineering and software development.

Technical Skills

Programming Languages: Python, SQL, Java, Angular
Framework & Libraries: Django, LangChain, Spring Boot, RESTful APIs, Pandas, NumPy
AI & GenAI: Amazon Bedrock, Hugging Face, OpenAI API, RAG, Prompt Engineering, LLM Orchestration, Fine-Tuning
Data & ETL: AWS Glue, AWS Athena
Cloud & Serverless: AWS Lambda, AWS S3, AWS EC2, AWS API Gateway, AWS SageMaker, AWS IAM, AWS CloudWatch
Vector Database: ChromaDB
Database: PostgreSQL, MySQL, MongoDB
Visualization & BI: Amazon QuickSight, Power BI
Tools: Git, Docker, Splunk, CI/CD Pipelines, SonarQube, Tricentis Tosca, GitHub Actions

Professional Experience

Pyramid Consulting | Software Engineer [Client: Nationwide Insurance] **January 2025 – June 2025**

- Implemented the Python based Analytical Model into the suitability application through Rest API ensuring seamless rule execution and decision scoring using the OpenL Tablets Library.
- Improved the Java suitability application's performance by refactoring components using the Factory Design Pattern and upgrading internal algorithms, resulting in a faster decision process and a 24% increase in annuity contract approval accuracy with OpenL Tablets Library.
- Scripted 20+ PostgreSQL queries for report generation and data analysis of the suitability product, improving decision-making and analyzing future enhancement capabilities.

Collabera | Python Developer [Client: Nationwide Insurance] **June 2022 – December 2024**

- **Python Project**
  - Designed an end-to-end AWS ETL pipeline where S3-triggered Lambda functions cleaned and transformed raw data in real time, ensuring consistent and analysis-ready outputs.
  - Cataloged transformed data using AWS Glue Crawlers, automatically generating schema definitions and enabling efficient querying through Athena.
  - Built structured analytical tables using Athena SQL and created 30+ AWS QuickSight datasets, supporting dashboards and market trend reporting for annuity contract insights.
  - Collaborated with cross-functional team in an agile environment using test driven development and leveraged agile methodologies to deliver project solutions integrating both front-end and backend components resulting in efficient product delivery and improved user experience.
  - Ran AWS Script using PowerShell to update reports on cloud dashboards.
- **Java Project**
  - Maintained Angular front-end for suitability application ensuring optimal user experience through regular updates and bug fixes resulting in improved user satisfaction and reduced error rates.
  - Upgraded the suitability application from Java 8 to Java 17, boosting application performance, strengthening security, and enabling compatibility with modern Spring Boot frameworks.
  - Integrated SonarQube into the development pipeline to automate code quality checks and performance scans, improving maintainability and achieving consistent 80%+ code coverage across services.
  - Developed Tosca scripts for end-to-end API testing and JUnit tests for Spring Boot microservices, improving test reliability and accelerating deployments by integrating them into the CI/CD pipeline.
  - Maintained application as an application support engineer to make application production ready and supported in every production deployment and live support during the deployment.

ConsultAdd Inc. | Software Developer **May 2021 – August 2022**

- Developed Python script and Lambda functions for data processing, including fetching data Elasticsearch, transforming Kinesis data and manipulating Salesforce data with Graph API call.
- Maintained and enhanced Master Data Management application with over 10,000 records and built test-driven development

pipelines using GitLab CI/CD improving development efficiency and data management.

- Developed and tested an OTP delivery API using Django Rest Framework and Plivo third party service, enhancing the login and authentication system with text messaging feature with 98% delivery success rate.

SUNY Binghamton | IT Assistant **July 2020 – April 2021**

- Applied growing skills in Python, full-stack development, and cloud concepts (AWS) while supporting internal IT tasks, helping with small automation scripts, and assisting in troubleshooting technical issues.
- Gained practical experience with web technologies and cloud-based tools by working on minor application enhancements, basic hosting tasks, and improving workflow efficiency within the department.

Certifications

- **AWS Certified Generative AI Developer – Professional |** Amazon Web Services March 2026 – March 2029
- **Microsoft Certified: Azure AI Fundamentals Certification |** Microsoft July 2025 – July 2026
- **AWS Certified Cloud Practitioner |** Amazon Web Services July 2021 – July 2024

Project Experience

AI Resume Analyzer

- Built an AI Resume Analyzer using spaCy NLP, LangChain, OpenAI, and Chroma vector DB to match resumes with job descriptions using semantic retrieval and structured LLM reasoning.
- Designed a RAG pipeline with OpenAI embeddings and Chroma vector database to retrieve the most relevant resume sections for accurate job-fit evaluation.
- Implemented NLP-based skill extraction and experience estimation using spaCy and fuzzy matching to improve resume parsing accuracy.
- Generated structured hiring reports (match score, missing skills, improvement bullets, tailored summary) using prompt-controlled LLM output with Pydantic validation.
- Developed a Streamlit web interface for resume upload, job description input, and transparent display of retrieved context and AI results.

Power BI Project – Data Professional Survey Analysis

- Leveraged Power Query to clean, transform and integrate raw data from Excel into structured data Model.
- Designed and developed an interactive Power BI dashboard to analyze a survey of 630 professionals across multiple roles.
- Visualized Key insights such as favorite programming language, country-wise distribution etc.
- Delivered data driven dashboard enabling comparative analysis of salaries, job roles and skill preferences.

Education

- **University of the Cumberlands** | Doctor of Philosophy (PhD) in IT-AI (Ongoing) May 2025 – Expected August 2028
- **Harrisburg University of Science & Technology |** Master of Science (MS) in Project Management May 2023 – April 2025
- **SUNY-Binghamton University |** Master of Science (MS) in Computer Science August 2018-May 2020
- **Amravati University |** Bachelor of Engineering (BS / BE) in Computer Science and Engineering July 2013- May 2017